Primary Mirror FEA Lightweighting Report

Structural & Opto-Mechanical Parameter Sensitivity Study

Siemens NX Pre/Post & NASTRAN
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OUTER DIAMETER (D)	RADIUS OF CURVATURE	CENTRAL SUPPORT HOLE	SUBSTRATE MATERIAL
560.0 mm	1085.0 mm (k = -1.22)	175.0 mm (Ri = 87.5 mm)	Zerodur (ρ = 2530 kg/m³)
OPTICAL DISH SAG	SUPPORT MOUNT	TARGET MASS	TARGET DISPLACEMENT
36.0 mm (f/0.97 fast)	18-Point Whiffletree	≤ 12.0 kg (61.6% lightened)	< 60.0 nm RMS / PV

1. Complete FEA Simulation & Parameter History

RUN ID / LABEL	DEPTH H (MM)	FACEPLATE TF (MM)	RIB TW (MM)	CELL L (MM)	FILLET RF (MM)	MASS (KG)	MAX DISPL. (NM)	ENGINEERING INSIGHT / NOTES
Point C	73.7	7.0	1.5	46.0	1.5	12.45	74.3	**BEST H=73.7** Balanced shear vs mass
Run 254	85.0	8.0	1.4	48.0	1.5	12.45	74.29	Star-node locking; main body < 40 nm
Run 209	73.7	9.0	1.4	48.0	1.5	13.03	75.38	Solid rib hubs (no through-holes)
Run 217	73.7	7.5	1.6	54.0	1.5	13.00	78.16	Continuous outer hoop ring restored
Run 251	85.0	8.0	1.4	48.0	1.5	12.30	80.23	Deep blank; slight row-snapping asymmetry
Run 221	73.7	7.0	1.8	52.0	1.5	12.60	80.70	Faceplate 7mm allowed local quilting
Point A	73.7	8.0	1.5	50.0	1.5	12.79	81.7	Moderate cell span with 8mm faceplate
Run 149	73.7	8.0	1.5	48.0	1.5	12.78	81.72	dist = 0.00 mm exact FE node mapping
Run 271	73.7	8.5	1.75	54.0	1.5	12.50	83.65	High faceplate rigidity; outer cantilever limit
Run 264	85.0	9.5	2.1	54.0	1.5	12.85	83.70	Body max 83.7 nm; **Optical face = 38–46 nm**
Run 215	73.7	7.0	1.8	54.0	1.5	12.95	83.88	Open perimeter notches caused saddle wave
Point B	73.7	7.5	1.2	42.0	1.5	12.19	86.7	**WORST** 1.2mm rib loss of shear modulus
Run 141	73.7	8.0	1.5	52.0	1.5	12.80	88.22	Non-snapped circle points on thin rib walls
Run 22	73.7	9.0	1.5	60.0	1.5	13.28	88.86	Initial coarse baseline (60 mm cell)
Run 269	73.7	7.0	1.5	48.0	1.5	12.398	90.89	NX Measured Volume = 4.90×10^6 mm³
Run 248	73.7	8.0	1.5	48.0	1.5	12.82	92.53	Outer hubs pulled inward → perimeter droop
Run 252	85.0	8.0	1.4	48.0	1.5	12.28	96.16	Points landed on knife-edge single rib edges
Run 226	73.7	7.5	1.35	46.0	1.5	12.10	103.10	Extreme thin-wall shear compliance

2. Key Opto-Mechanical Takeaways & Design Rules

1. The Rib Thickness vs. Shear Modulus Rule (Point B vs. Point C)

- **Why Point B was the worst (86.7 nm) despite smallest cell (L = 42 mm):** Thinning the rib below 1.5 mm (tw = 1.2 mm) causes a catastrophic drop in web shear modulus. An open-back mirror under 1g gravity relies heavily on rib shear stiffness to transmit loads to support pads. Small cell size cannot compensate for a flimsy 1.2 mm rib.
- **The Minimum Rib Rule:** For a 560 mm Zerodur primary mirror, rib thickness must stay **tw ≥ 1.5–1.8 mm**.

2. 6-Rib Star Node Anchoring vs. Knife-Edge Single-Rib Contacts

- When support points land on a single 1.4 mm thin rib wall or near open pocket cavities (Run 252), local knife-edge compliance adds over **+22 nm** of artificial deflection.
- Anchoring all 18 Whiffletree points **dead-center on solid 6-rib star node vertices** (Point C, Run 254) distributes reaction loads symmetrically in 6 directions, dropping deflection immediately to **74 nm**.

3. Optical Surface Sag vs. Full-Body Maximum (Run 264)

- In deep open-back mirrors (H = 85 mm), the numerical full-body max (83.7 nm) occurs at the unsupported lower tips of the back ribs.
- The **front optical reflective faceplate** achieves an optical surface sag of **38.0–46.5 nm**, successfully satisfying the < 60 nm optical requirement!